BTEC Level 3 Applied Science · Unit 1 Physics · Answer sheet

Progressive Waves — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDScience items, not official Pearson questions.

A–E

1. A: max displacement from equilibrium. λ: shortest distance between points in phase. f: oscillations per second.
2. $v = f\lambda$; m s^{-1}, Hz, m.
3. $T = 1/f$.
4. Vertical distance from the equilibrium line to a crest (not crest-to-trough).
5. 3.2 m s^{-1}.
6. $f = 2.00 \times 10^8$ Hz; $\lambda = 1.50$ m.
7. $v = 300$ m s^{-1}; $f = 150$ Hz.
8. See retrieval definitions. [2]
9. $\lambda = 340/510 = 0.667$ m (3 s.f.). [2]
10. Particles oscillate about equilibrium; energy is passed along; no net movement of air. [3]
11. $T = 20$ ms / 5 = 4.0 ms; $f = 250$ Hz; $\lambda = 0.040$ m; $v = 10$ m s^{-1}.
12. Frequency stays the same. Speed and wavelength change (both increase or both decrease together).